Bachelor Thesis

Proactive AI moderation for goal-oriented remote meetings: Design and acceptance of a context-sensitive system to support convergent decision-making processes

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Contents

1 Introduction

Remote meetings have become a cornerstone of modern collaboration. Since the start of the COVID-19 pandemic, the use of video conferencing (VC) industry has risen dramatically [**iyengarZoomsRevenueSoars2020**]. Hardly any meeting takes place without at least one remote participant [Source] Yet they often suffer from inefficiencies, particularly in convergent decision-making processes such as prioritization, problem-solving, and consensus-building. Research by Atlassian indicates that up to 72% of meetings are considered ineffective, with common issues including off-topic discussions, unequal participation, and a lack of structure. [1]

Although traditional facilitation strategies are effective, their practical implementation is limited by the fact that not every team has permanent access to a trained facilitator. Professional training and external consulting, while valuable, are costly and not continuously available.

The potential of AI-driven facilitation to address these gaps lies in its ability to provide real-time, context-sensitive support. Unlike static tools (e.g., timers or agendas), proactive AI systems can analyze conversations, detect deviations from predefined goals, and intervene with constructive suggestions—thereby enhancing goal orientation and reducing cognitive load on human moderators. Early adopters of AI in meetings, such as Otter.ai (real-time transcription) and Microsoft Teams' AI recaps, demonstrate the feasibility of automated support, but these tools lack proactive, adaptive moderation tailored to convergent processes [Microsoft, 2024; Otter.ai, 2023]

1.1 Background and Motivation

1.1.1 Inefficiencies in remote meetings: Challenges in convergent decision-making processes

Convergent decision-making in remote meetings faces unique obstacles:

- Off-topic discussions: Studies show that 30–50% of meeting time is spent on irrelevant tangents, derailing progress [Rogelberg et al., 2006]. [2]
- Unequal participation: Dominant voices can overshadow quieter team members, leading to groupthink and suboptimal outcomes [Nunamaker et al., 1991].
- Lack of structure: Without clear facilitation, meetings often lack timeboxing, role assignment, or explicit decision criteria, reducing efficiency [Schwartz, 2016].

Traditional solutions—such as hiring professional facilitators or training team members—are cost-prohibitive or scalability-limited. For example, a certified facilitator may charge €100–300/hour [International Association of Facilitators, 2024], making continuous support impractical for most organizations. Furthermore, not every team has permanent access to trained personnel, and ad-hoc facilitation often fails to address systemic issues like dysfunctional team dynamics or blind spots in group perception [Edmondson, 1999].

1.1.2 Potential of AI-driven facilitation to enhance goal orientation

AI can bridge this gap by:

- Automating real-time analysis: Natural Language Processing (NLP) can transcribe and semantically analyze discussions to identify deviations from the meeting's goal [Bommasani et al., 2021].
- Providing adaptive interventions: Context-sensitive prompts (e.g., "Let's return to the agenda: prioritizing the top 3 solutions") can refocus conversations without human intervention.
- Democratizing participation: AI can nudge underrepresented voices (e.g., "Alex, you haven't shared your perspective yet—would you like to add anything?") or summarize key points to ensure alignment.

Pilot studies on AI moderation tools (e.g., Humley's AI meeting assistant) report 20–30% reductions in meeting duration and improved participant satisfaction [Humley, 2023]. However, these tools often lack proactivity—reacting to issues rather than preventing them—and customization for specific use cases like convergent decision-making.

1.2 Research Problem and Gap

1.2.1 Limitations of existing moderation tools in remote settings

Current tools fall short in three key areas:

- Passive functionality: Most AI tools (e.g., Zoom AI Companion, Google Meet's recaps) focus on post-meeting summaries or transcription, not real-time moderation [Zoom, 2024; Google, 2023].
- Generic interventions: Solutions like Miro's AI facilitator provide template-based guidance but fail to adapt to the meeting's context or user preferences [Miro, 2024].
- Lack of empirical validation: Few studies evaluate user acceptance of proactive AI moderation, particularly in high-stakes decision-making scenarios [Benbasat and Barki, 2007].

1.2.2 Need for context-sensitive, proactive AI systems

To address these gaps, this thesis proposes a proactive AI moderation system that:

- Detects thematic deviations in real time using NLP and meeting goal tracking.
- Intervenes constructively with personalized, non-intrusive prompts (e.g., private messages to individuals or group reminders).
- Validates design choices through user studies focusing on usability (SUS) and acceptance (TAM).

Such a system could reduce meeting fatigue, improve decision quality, and lower the barrier to effective facilitation—but its success hinges on user trust and perceived usefulness, which remain underexplored in literature.

1.3 Research Objectives and Questions

This thesis addresses the inefficiencies in goal-oriented remote meetings through a structured approach that integrates three complementary research objectives. The first objective (RQ1) investigates facilitating strategies and challenges in remote meetings to establish a foundation based on real-world practices and pain points. The second objective (RQ2) focuses on designing an AI-based moderation system that translates these insights into technical requirements, ensuring the system is context-aware and goal-oriented. The third objective (RQ3) evaluates user acceptance among knowledge workers.

The work bridges the gap between theoretical facilitation techniques and practical AI-driven solutions, ensuring that the system is both technically robust and user-centric.

1.3.1 RQ1: Facilitating strategies and challenges in goal-oriented remote meetings

> *Which facilitating strategies are suitable for the goal-oriented management of remote meetings with convergent objectives (such as decision-making, prioritization, problem-solving)? What specific challenges typically arise in this context?*

The first research question aims to identify effective strategies for managing convergent remote meetings, such as decision-making, prioritization, and problem-solving, while also analyzing the specific challenges that arise. To achieve this, the thesis conducts expert interviews with certified facilitators, agile coaches, and team leads to gather empirical insights. It also analyzes literature on facilitation techniques, including timeboxing and role assignment, as well as the dynamics of remote collaboration. The findings are compared with existing research to derive best practices and identify pain points, such as off-topic discussions and unequal participation, which AI could potentially address.

1.3.2 RQ2: Design requirements for an AI-based moderation system

> *How should an AI-based moderation system be designed that autonomously detects thematic deviations based on a predefined meeting goal and real-time transcript analysis and promotes goal orientation through constructive, context-sensitive interventions?*

The second research question focuses on developing a proactive AI system that detects thematic deviations in real time and intervenes contextually to maintain goal orientation. This objective is addressed by defining functional requirements, such as real-time transcript analysis and adaptive interventions, and non-functional requirements, including privacy, latency, and scalability. A system architecture is then designed with components for transcription, NLP-based analysis, and intervention logic, incorporating mechanisms for private nudges or group reminders. A prototype is implemented using NLP libraries like Hugging Face Transformers, and its technical feasibility is evaluated.

1.3.3 RQ3: User acceptance among knowledge workers

> *How do knowledge workers in remote teams evaluate the usefulness, usability, and psychological effects of such an AI moderation tool, and what factors influence its acceptance?*

The third research question assesses how knowledge workers perceive the usefulness, usability, and psychological impact of the AI moderation system, as well as the factors influencing its adoption. A usability study is conducted with remote teams, using frameworks such as the System Usability Scale and the Technology Acceptance Model. Quantitative data, including SUS scores and efficiency metrics, are collected alongside qualitative feedback from interviews that explore perceptions of control, trust, and pressure. The analysis also examines psychological effects, such as relief versus intrusion, and adoption barriers, including resistance to automation and privacy concerns.

2 Theoretical Foundations

This chapter presents the theoretical foundations and the current state of research. These form the basis for the subsequent work. Based on a systematic literature review, five key challenges in the context of meetings were identified (2.1.3). These serve as the starting point for the in-depth analysis in the following expert interviews (see Chapter 3) and provide indicators of what to focus on during the subsequent requirements gathering process (see Chapter 4.1).

2.1 Remote Meetings and Convergent Decision-Making

2.1.1 Meetings in Modern Work: Trends and Context

The Doolde Report "State of Meetings 2023" finds that 40% of meetings are scheduled 1–4 weeks in advance (55% in North America). In contrast, the Mirsooft report reveals that most meetings (57%) are ad hoc, without formal invitations. [**bobbyDoodleStateMeetings2026**]

2.1.2 Definitions and distinctions (convergent vs. divergent processes)

2.1.3 Key challenges: Off-topic discussions, unequal participation, lack of structure

A clear issue with goal-oriented meetings is the absence or unclear communication of objectives. It seems plausible that a significant number of meetings are scheduled without a defined goal or agenda (as discussed in section 2.1.1). A lack of structure in meetings is therefore one of the key challenges. While an agenda can serve as the foundation for such structure, it alone does not guarantee a successful meeting. Meeting goals are often dynamic and may need to be adjusted as the meeting progresses, due to the associated uncertainties [2, chap 4.1]. Although the agenda can outline a path to achieving the overall meeting goal [**scottMentalModelsMeeting2024**], without linked sub-goals, it guarantees neither the success of the meeting nor a timely conclusion. [2, chap 4.1] Discussions can nevertheless get out of hand if participants do no know what to priorities. [2, chap 4.1] An agenda was identified by many respondents (67%) as a key element for successful meetings in Doodle's State of Meetings Report 2019. Only the setting of clear meeting objectives (72%) was cited even more frequently. [**DoodleStateMeetings2019**]

Based on this, the following summary challenge statement was derived:

> **Challenge 1: Clear goal-setting and structure.** Goal-oriented meetings frequently lack clearly communicated objectives and a reliable structure, so that discussions lose focus and the meeting neither succeeds nor concludes on time.

Chen et al. noted, as a secondary finding, that individuals in higher hierarchical positions tended to intervene more frequently and maintain focus on meeting objectives when a clear list of goals was provided. Conversely, those in lower hierarchical positions felt less obligated to intervene, even when they observed deviations from the main topic. [2] Doodle's Stat of Meetings Report 2019 identified a correlation between salary and self-perceived participation in meetings. Respondents with higher salaries were more likely to describe themselves as active participants. [**DoodleStateMeetings2019**] This may stem from inexperience, hierarchical constraints, group dynamics, or a lack of error culture, where dissent risks loss of face. Low participation can narrow solution spaces, yield monotonous outcomes, and reduce meeting effectiveness.

Dysfunctional communication—such as complaining, criticizing, and digressing—reduces effectiveness. [**kauffeldMeetingsMatterEffects2012**]

-> Unausgeglichen / geringe participation -> Unvorhergesehen veränderungen im Meeting (flasch Annahmen, meetingziel ändert sich) -> technische Probleme: Instablie internetverbindungen, Störungen etc. (nicht im scope dieser arbeit aber dennoch relevant)

2.2 Facilitation Strategies for Remote Collaboration (RQ1)

2.2.1 Empirical evidence on effective techniques (e.g., timeboxing, role assignment)

2.2.2 Technological and social dynamics in virtual teams

2.3 AI in Meeting Support Systems (RQ2)

2.3.1 AI-Agents in General

zunächste soll geklrät werden was unter KI-Agenten verstanden wird. Letztlich sind KI-Agenten system um teilaufgaben zu automaiseren. Um eine präzieres Verständnis zu bekommen lohtn es sich daher eine abgrezung zu klasssen worklof automatiserungen zu machen.

Programmatische Prozessautomatisierung bezeichnet die regelbasierte Automatisierung von Abläufen, bei der festgelegte Bedingungen und strukturierte Entscheidungslogiken genutzt werden, um Prozesse effizient, zuverlässig und wiederholbar auszuführen. Technicken die hier zum einsatz kommen sind robotic Process Automation, Skritping und Markos.

LLMs können kontext ldabei helfen unstruktierte aufgaben zu lösen. antrhopic whält folgende Defintion bz von Agenten welche KI-Agneten von Workflos abgrenzt. Workflows are systems where LLMs and tools are orchestrated through predefined code paths. Agents, on the other hand, are systems where LLMs dynamically direct their own processes and tool usage, maintaining control over how they accomplish tasks.

2.3.2 Building blocks for AI-Agents

Augmented LLM

Prompt chaining

Routing

Parallelization

Orchestrator-workers

Evaluator-optimizer

2.3.3 Frameworks to design LLM-based Meeting AI-Agents

The Observe, Ask, Intervene Framework for designing AI Agents for more inclusive Meetings

In study by Houtti et al.[3, p. 5], in which participants were asked to design virtual co-hosts during semi-structured design sessions, a widespread tendency emerged to design the co-host to be interactive rather than to let it act entirely automatically. Common design patterns involved the co-host providing prompts in response to explicit instructions (e.g., when explicitly asked or when a pin was set). The authors conclude that interactivity could help prevent mistrust.

What? When? Who? Framework for Goal relecting (Are we on track?)

AI-Agents in General

2.3.4 Proactive AI-Interactions: Timing and triggers for Human-AI-Interaction-Systems

To reduce the need for explicit prompts, an AI agent should be able to proactively recognize when it is its turn in an interaction sequence. In dyadic human-AI interactions, pauses or so-called “stop words” are often used for this purpose to identify a change in speaker. In multi-party scenarios (e.g., group conversations), however, this becomes significantly more complex and leads to the “multi-party turn-taking problem”. [**sacksSimplestSystematicsOrganization1974**]

Some approaches model the problem as a machine learning task by attempting to predict the next speaker based on previous speaker changes. [**bayserLearningMultiPartyTurnTaking2019**] Bayer et al. employ traditional algorithms such as Maximum Likelihood Estimation (MLE) and Support Vector Machines (SVMs), as well as deep learning architectures including Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks. They distinguish two primary paradigms: agent-only models, which rely solely on speaker history, and agent-and-content models, which additionally incorporate conversational context. Both paradigms have been trained and evaluated across diverse, topic-oriented and chit-chat corpora using these varied algorithms. The empirical findings indicate that while simple agent-only models are sufficient for small, highly focused dialogues, incorporating conversational content becomes crucial in goal- and topic-oriented interactions characterized by complex, irregular turn-taking patterns. Conversely, in unstructured chit-chat scenarios, the integration of semantic content yields no statistically significant performance gain over simple, rule-based baselines such as the "Repeat Last heuristic" [**bayserLearningMultiPartyTurnTaking2019**]

With these approaches, however, it remains unclear why an agent should interact at a specific point in time. Due to the design of the ML algorithms, the timing of the interaction is decoupled from its cause or is only implicitly included in the model, but cannot be explicitly explained. (Example: We know that it is now the time for the agent to interact now -> But why? Whats the reason its like that he should talk? What should he say?)

Die über die leltzten jahre stattgefunden entwicklung bei LLMs ermöglichen neue ansätze um diese Problem zu modellieren. Neus Paradgima von LLM-basierten Agentischen Systemen. Was ist das? -> Hier berscheiben was KI-Agenten sind (Anthopic definition finden und verwenden) LLM basierte AI-Agenten mit einer LLM pipelie welche intrisische gedanken simmuliert, könnte ein viel versprechender ansatz sein.

Liu et al. verfolgenden diesen Ansatz um einen procativen conversationa Agent mit inner Thoughts zu modellieren. [4]

Y verfolgen diesen ansatz um eine proaktiven Chatbot zu modellieren und schlussfolgern, dass

Classic Worksflow

2.3.5 Theoretical frameworks: Human-AI Interaction (HAI), Explainable AI (XAI)

2.4 Technology Acceptance and Usability

2.4.1 Technology Acceptance Model (TAM)

2.4.2 System Usability Scale (SUS) and psychological effects

3 Empirical Analysis of Facilitating Strategies (RQ1)

3.1 Methodology: Expert Interviews

3.1.1 Sampling: Purposeful selection of 5 facilitating experts

3.1.2 Data Collection: Semi-structured interviews (30 min)

3.1.3 Analysis: Thematic coding

3.2 Results: Facilitating Strategies and Challenges

3.2.1 Comparison with literature (Chapter 2)

3.3 Implications for AI System Design

3.3.1 Derived requirements for RQ2

4 System Design and Implementation (RQ2)

4.1 Requirements Analysis

4.1.1 Functional Requirements

-> es gibt zwei hautp unterschieden die man treffen kann: procativ system und reactive systeme. In are we on track wurden zwei verscheidene system design (aktiv und passiv) beiee hatte vor und nachteile. (Welche) Sind aber beide relevant. Daher Plattform, welche beides kann, agenda live verfolgen und agent kann selbst einschreiten.

4.1.2 Non-Functional Requirements

4.2 System Architecture

4.2.1 Components: Transcription, Analysis, and Intervention Modules

4.2.2 Technology Stack: NLP Libraries, Backend/Frontend Frameworks

4.3 Prototype Development

4.3.1 Implementation Details

4.3.2 Limitations of the Prototype

5 Empirical Validation (RQ2 and RQ3)

5.1 Methodology: Usability Study

5.1.1 Study Design

5.1.2 Sample

5.1.3 Data Collection

5.2 Results

5.2.1 Usability (SUS Scores and Analysis)

5.2.2 Acceptance (TAM: Perceived Usefulness, Ease of Use, Psychological Effects)

5.3 Discussion

5.3.1 Implications for System Design

5.3.2 Theoretical Contributions

5.3.3 Practical Impact for Organizations

Bibliography

[1] Atlassian. *Workplace Woes: Meetings*. Inside Atlassian. URL: `https://www.atlassian.com/blog/workplace-woes-meetings` (visited on 08/26/2026).

[2] Xinyue Chen et al. "Are We On Track? AI-Assisted Active and Passive Goal Reflection During Meetings". In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. CHI 2025: CHI Conference on Human Factors in Computing Systems. Yokohama Japan: ACM, Apr. 26, 2025, pp. 1–22. ISBN: 979-8-4007-1394-1. DOI: `10.1145/3706598.3714052`. URL: `https://dl.acm.org/doi/10.1145/3706598.3714052` (visited on 02/26/2026).

[3] Mo Houtti et al. "Observe, Ask, Intervene: Designing AI Agents for More Inclusive Meetings". In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. CHI 2025: CHI Conference on Human Factors in Computing Systems. Yokohama Japan: ACM, Apr. 26, 2025, pp. 1–18. ISBN: 979-8-4007-1394-1. DOI: `10.1145/3706598.3713838`. URL: `https://dl.acm.org/doi/10.1145/3706598.3713838` (visited on 02/24/2026).

[4] Xingyu Bruce Liu et al. "Proactive Conversational Agents with Inner Thoughts". In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. CHI 2025: CHI Conference on Human Factors in Computing Systems. Yokohama Japan: ACM, Apr. 26, 2025, pp. 1–19. ISBN: 979-8-4007-1394-1. DOI: `10.1145/3706598.3713760`. URL: `https://dl.acm.org/doi/10.1145/3706598.3713760` (visited on 02/24/2026).

6 Appendix

6.1 Semi-Structured Questionnaire for Interviews with Facilitators

Background and Role

1.1 How do you feel about online collaboration in general?

1.2 To start with facilitation: do you hold a generally positive or critical perspective toward online facilitation? Why?

1.3 How long have you been working as a facilitator?

1.4 Why did you start facilitating, or how did you end up becoming a facilitator?

1.5 What do you like about being a facilitator?

Online vs. In-Person Collaboration

2.1 Compared with in-person collaboration, how would you rate online collaboration overall?

2.2 In your experience, what are the main differences between online and in-person collaboration?

Facilitation Experience and Context

3.1 Can you describe your facilitation experience in online collaboration settings?

3.2 Who were the participants in the sessions you facilitated?

3.3 What were the key points you focused on when facilitating this type of event?

3.4 What were the main difficulties or challenges you encountered as a facilitator?

Platforms and Tools

3.1 What platforms or tools did you use during facilitation?

3.2 From a facilitation perspective, how did these platforms help you?

3.3 How do you expect platforms or tools could better support you as a facilitator in the future?

Facilitation Style

3.1 Wann sollte jemand direkt in der Gruppe angesprochen werden? Wann lieber privat?

3.2 Was sind typsiche kritishce Momente in ziel-orienterne Meeting in denen eine entscheidung getroffen werden soll? Was könnte schief gehen?

3.3 3.1 Woran erkennst du diese Momente? Kannst du dies quantifizieren?

3.4 Welche Rolle spielt die Vorbereitung auf das Meeting?

3.5

3.6 Wie stellst du sicher, dass das Meeting on track bleibt?

  3.1 Welche speziellen Methoden / Strukturen / Angewohnheiten, haben sich für dich bewährt?

  3.2 Wie stellst du sicher, dass den Teilnehmende klar ist was ”on Track” bedeutet bzw. was das aktuelle das Ziel ist?

Designen an AI Facilitator

3.1 Wie sollte ein AI Facilitator aussehen??

3.2 Was würdest du als essentielle Features beschreiben??

3.3 Wo siehst du möglice schwächen eines AI-Systems, welche proactiv in Meetings einschreitet?